

PowerCell PC-42

Low-Dropout Linear Regulator - Datasheet Rev C

The PowerCell PC-42 is a low-dropout linear regulator designed for noise-sensitive analog and mixed-signal rails in industrial control equipment. It accepts a wide 6 V to 24 V input and delivers a fixed 3.3 V output at up to 1.5 A continuous. Dropout voltage is 310 mV at full load. The device integrates thermal shutdown, current limiting, and reverse-polarity protection.

Because the PC-42 is a linear topology, power dissipation is the input-to-output differential multiplied by load current. At 24 V input and 1.5 A load the device dissipates 31 W, which exceeds the package rating without a heatsink; the recommended maximum input for continuous full-load operation is 12 V.

Table 1: Absolute Maximum Ratings

Parameter	Value
Input voltage range	6 V to 24 V
Output voltage	3.3 V fixed
Continuous output current	1.5 A
Dropout voltage at 1.5 A	310 mV
Thermal shutdown threshold	150 C
Quiescent current	45 uA
Operating temperature range	-40 C to +125 C

Figure 1: PC-42 power subsystem block diagram

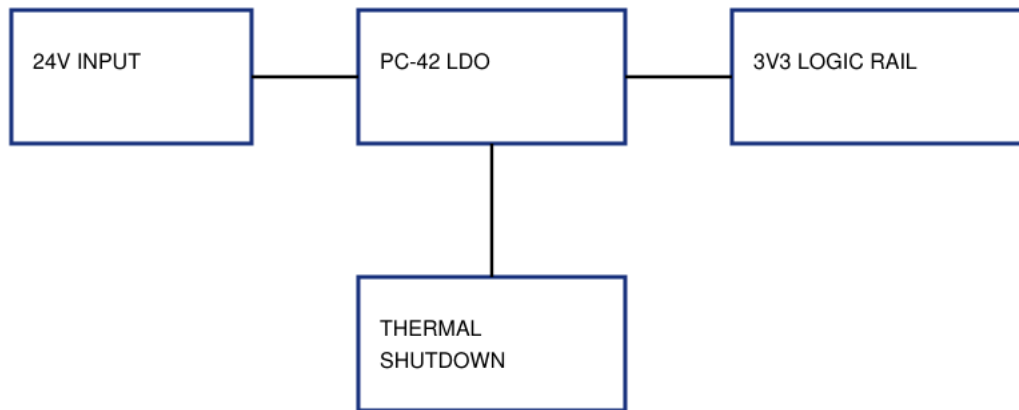


Figure 1: PC-42 power subsystem block diagram

The 24 V bulk rail enters through a reverse-polarity FET and is regulated down to the 3V3 logic rail. The thermal shutdown block monitors junction temperature and disables the pass element at 150 C, re-enabling with 15 C of hysteresis.

Efficiency at nominal 12 V input and 3.3 V output is 27 percent, which is inherent to linear regulation and not a defect of this part.

Application Notes

Output capacitor selection: the PC-42 requires a minimum 10 uF ceramic output capacitor with ESR below 100 mOhm for stability. X7R dielectric is recommended over Y5V because of Y5V's bias-voltage derating.

Layout: keep the feedback trace away from the switching node of any upstream converter. The exposed pad must be tied to the ground plane with at least nine thermal vias.

Qualification: the PC-42 is AEC-Q100 Grade 1 qualified and carries a 10-year supply commitment from the date of Rev C release.